

Result:



Neptun ID:

Draft papers: ☐

Name: _____

The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | (essay) | | | |
| 2.) | (essay) | | | |
| 3.) | ☐ | ☐ | ☐ | ☐ |
| 4.) | ☐ | ☐ | ☐ | ☐ |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the interpretation of a zero-order language!

2.) Define the conjunctive normal form (CNF).

3.) Which statements are true for the following three formula?

$$A \rightleftharpoons q \vee q, \quad B \rightleftharpoons q \vee \neg q, \quad C \rightleftharpoons \neg p \supset (q \supset p)$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

-1 $A, B \models C,$

2 $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \rightleftharpoons q \supset q, \quad B \rightleftharpoons \neg q \supset q \wedge p, \quad C \rightleftharpoons q \supset \neg p$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

-1 $A, B \models C,$

2 $(A \vee B)$ is valid.

Result:



Neptun ID:

Draft papers: ☐

Name: _____

The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | (essay) | | | |
| 2.) | (essay) | | | |
| 3.) | ☐ | ☐ | ☐ | ☐ |
| 4.) | ☐ | ☐ | ☐ | ☐ |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) What formula labels the root of the construction tree?

2.) Define the elementary conjunction.

3.) Which statements are true for the following three formula?

$$A \Rightarrow \neg q \supset q, \quad B \Rightarrow \neg q, \quad C \Rightarrow (p \equiv p) \supset q \vee q$$

-1 $A \Leftrightarrow B,$

-1 $\{A, B, C\}$ is satisfiable,

2 $A, B \models C,$

2 $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Rightarrow q \equiv q, \quad B \Rightarrow \neg(q \equiv p), \quad C \Rightarrow q \supset \neg(q \vee p)$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

-1 $A, B \models C,$

2 $(A \vee B)$ is valid.

Result:



Neptun ID:

Draft papers: ☐

Name: _____

The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | (essay) | | | |
| 2.) | (essay) | | | |
| 3.) | ☐ | ☐ | ☐ | ☐ |
| 4.) | ☐ | ☐ | ☐ | ☐ |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the semantic value of $(A \equiv B)$!

2.) Define the literal.

3.) Which statements are true for the following three formula?

$$A \Rightarrow q \equiv p, \quad B \Rightarrow \neg q, \quad C \Rightarrow (p \wedge q) \wedge q \vee \neg q$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

2 $A, B \models C,$

-1 $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Rightarrow p \supset q, \quad B \Rightarrow \neg(p \wedge q), \quad C \Rightarrow \neg q \wedge (q \supset q)$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

-1 $A, B \models C,$

2 $(A \vee B)$ is valid.